Assignment-4.R

ganga

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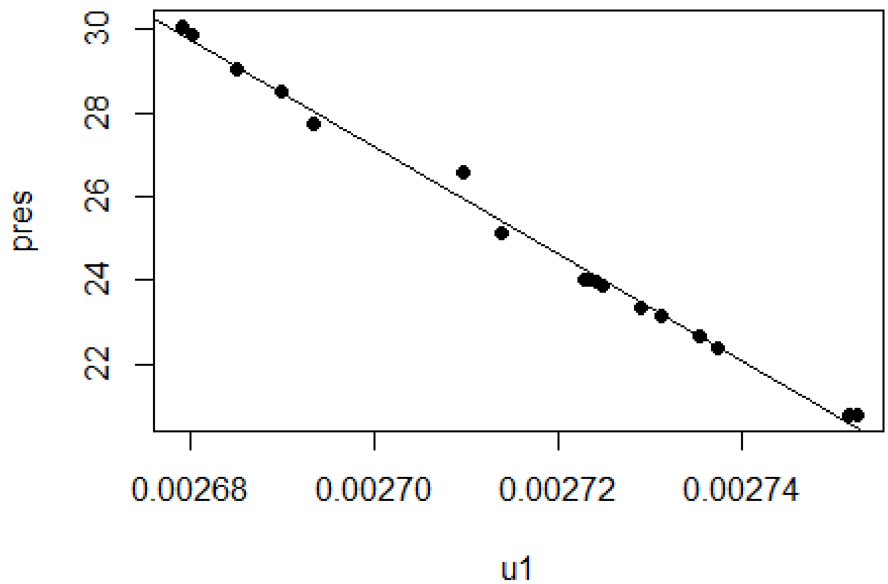
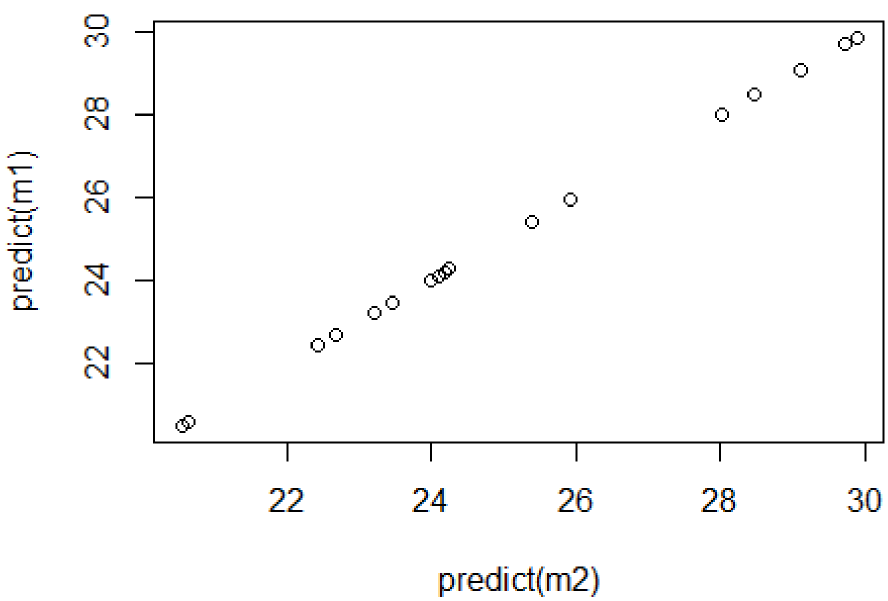
```
# ASSIGNMENT-4
# PROBLEM-2.7
#2.7.1
library(alr4)

## Loading required package: car

## Loading required package: carData

## Loading required package: effects

## lattice theme set by effectsTheme()
## See ?effectsTheme for details.

# Added a variable 'u1' by using the formula and the values given in the
question.
Forbes$u1 <- 1/(255.37 +(5/9)*Forbes$bp)
# Plotting the pres vs u1
with(Forbes, plot(pres~u1,pch=19))
with(Forbes, abline(lm(pres~u1)))
```

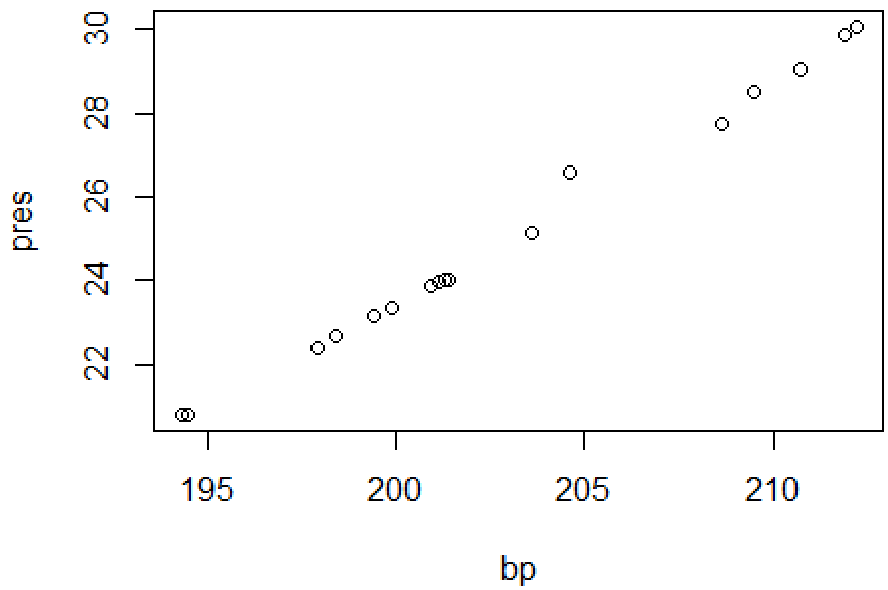



```
# (The 16 observations in the data is close to the staright line apart from
# the 12th observation which is far from the staright line).
# (The slpoe in the fig 1.4(a) is positive and here the slope is negative
because
# of the inverse equation of bpkelvin)

#2.7.2
# Given equation is 'E(pres/bp)=β0+β1u1' lets summarize the results
m1 <-lm(pres~u1,Forbes)
summary(m1)

##
## Call:
## lm(formula = pres ~ u1, data = Forbes)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.28216 -0.12643 -0.05569  0.17111  0.62569
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.723e+02  7.013e+00   53.08   <2e-16 ***
## u1          -1.278e+05  2.581e+03  -49.51   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2433 on 15 degrees of freedom
```

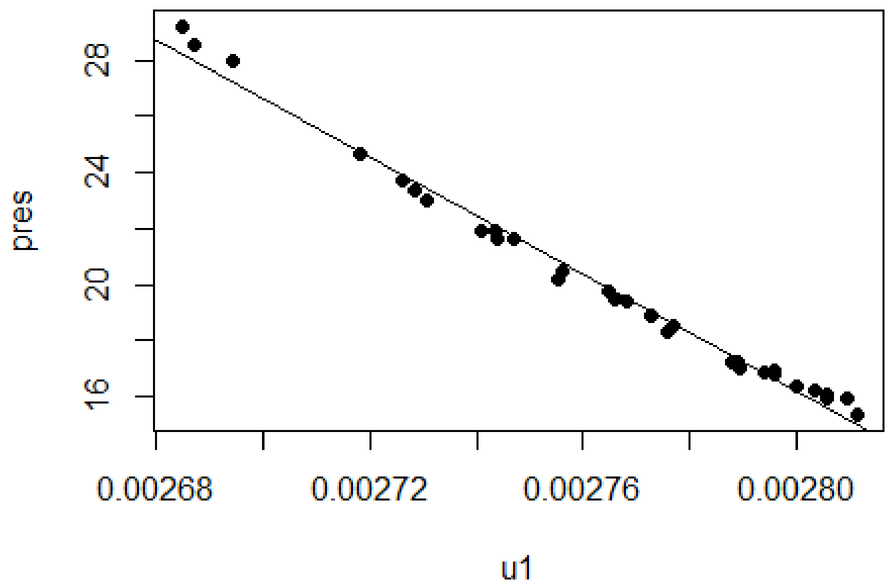
```
## Multiple R-squared:  0.9939, Adjusted R-squared:  0.9935
## F-statistic:  2451 on 1 and 15 DF,  p-value: < 2.2e-16

# (The result that we got is in the scientific form beacuse the data ot u1 is
# in decimal form and the coef of determination is 99% which shows residuals
# are near to mean line other than 12th observation)

#2.7.3
# Now we have 2 plots predictor as bp and as u1
m2 <- lm(pres~bp,Forbes)
plot(predict(m1)~predict(m2))
```




```
# The predictions of the two lines is equal(Y=X) and fit.
#Clarifying through our prediction model
plot(pres~bp,Forbes)
```

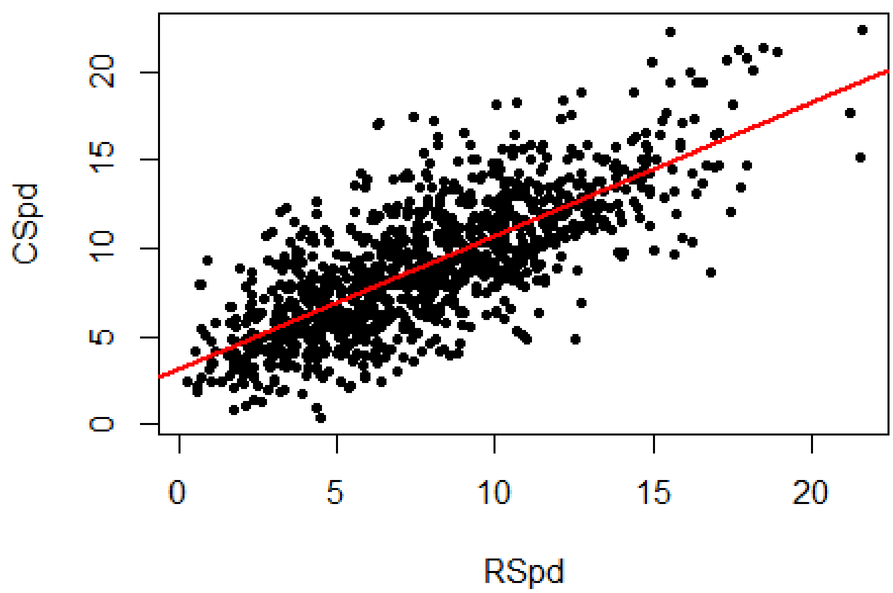



```
predict(m2, newdata=data.frame(bp=c(195)),interval="prediction")

##        fit      lwr      upr
## 1 20.90029 20.36167 21.43891

predict(m1, newdata=data.frame(u1=c(0.00275)),interval="prediction")

##       fit      lwr      upr
## 1 20.8011 20.23698 21.36522

# The prediction of the fit values are almost equal with very minimal
deviation

#2.7.4
# lets load the data collected by the botanist Joseph D. Hooker
# Added a variable 'u1' by using the formula same as before.
Hooker$u1 <- 1/(255.37 +(5/9)*Hooker$bp)
with(Hooker, plot(pres~u1,pch=19))
with(Hooker, abline(lm(pres~u1)))
```



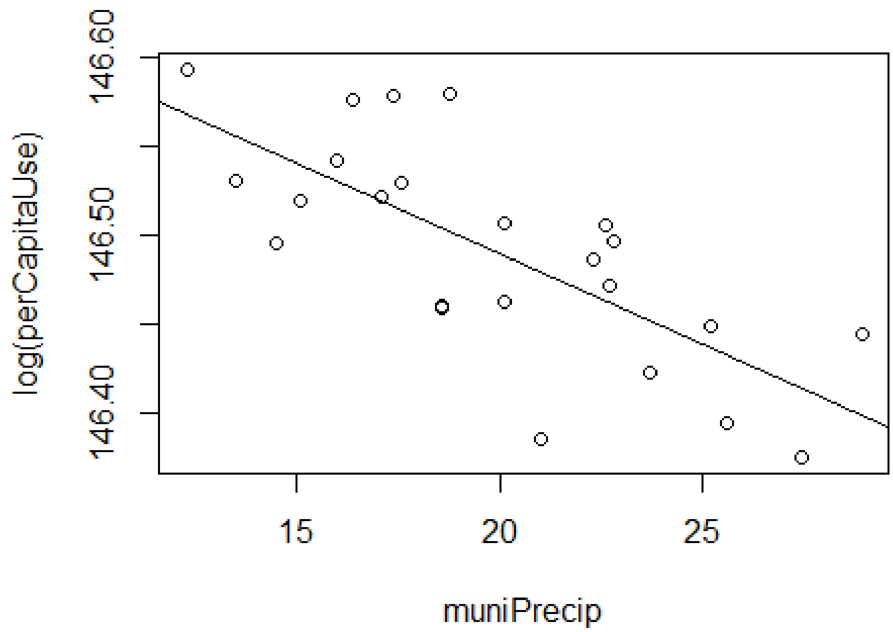

```
m3 <- lm(pres~u1,Hooker)
summary(m3)

##
## Call:
## lm(formula = pres ~ u1, data = Hooker)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.6671 -0.2751 -0.1478  0.3161  0.9427
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.089e+02  5.559e+00   55.57   <2e-16 ***
## u1          -1.045e+05  2.011e+03  -51.97   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.405 on 29 degrees of freedom
## Multiple R-squared:  0.9894, Adjusted R-squared:  0.989
## F-statistic:  2701 on 1 and 29 DF,  p-value: < 2.2e-16

# (This mean function is also looks alike Forbes and the mean is
fit.according to
# the summary multiple R-sqrd is 98% where residuals are near fit.

#problem-2.8
```

```r
#(Here given that α =β0+β1xbar where the value of α depends upon the value of mean
# which means that the value of mean doesn't change with each observation of x(predictor))

#Problem-2.21
#2.21.1
with(wm1, plot(CSpd~RSpd,pch=20))
with(wm1, abline(lm(CSpd~RSpd),col="red",lwd=2))
```




```r
#(According to the plot the response CSpd and the predictor RSpd are correlated
# and it has constant variance.So, the linearity is pausible for this data)

#2.21.2
m4<-lm(CSpd~RSpd,wm1)
summary(m4)

##
## Call:
## lm(formula = CSpd ~ RSpd, data = wm1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.7877 -1.5864 -0.1994  1.4403  9.1738
##
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.14123    0.16958   18.52   <2e-16 ***
## RSpd         0.75573    0.01963   38.50   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.466 on 1114 degrees of freedom
## Multiple R-squared:  0.5709, Adjusted R-squared:  0.5705
## F-statistic:  1482 on 1 and 1114 DF,  p-value: < 2.2e-16

#(Summary shows that intercept=3.14123,slope=0.75573,multiple R sqrd shows
that
# 57% of variability in CSpd is explained by RSpd.Hence the regression may
not be
# be good explaining the relationship of variables)

#2.21.3
# Lets obtain a 95% prediction interval for CSpd at a time when RSpd =
7.4285.
predict(m4, newdata=data.frame(RSpd=c(7.4285)),interval="prediction",
level=0.95)

##        fit      lwr      upr
## 1 8.755197 3.914023 13.59637

# Problem-4.13
# According to the formula log(perCapitaUse) = log(10^6muniUse/muniPop) we
add a variable
# because we have to show it as a response variable
MinnWater$perCapitaUse <- with(MinnWater,10^68*muniUse/muniPop)
m5<-lm(log(perCapitaUse)~year+muniPrecip,MinnWater)

require(rgl)

## Loading required package: rgl

with(MinnWater, plot3d(log(perCapitaUse)~year+muniPrecip, type = "s", col =
"red", size = 1))

plot(log(perCapitaUse)~muniPrecip,MinnWater)

abline(lm(log(perCapitaUse)~muniPrecip,MinnWater))
```




```
summary(m5)

##
## Call:
## lm(formula = log(perCapitaUse) ~ year + muniPrecip, data = MinnWater)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.09766 -0.03057  0.01086  0.02871  0.07577
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.463e+02  2.593e+00  56.417  < 2e-16 ***
## year         2.155e-04  1.297e-03   0.166     0.87
## muniPrecip  -1.026e-02  2.084e-03  -4.922 7.21e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.04397 on 21 degrees of freedom
## Multiple R-squared:  0.5356, Adjusted R-squared:  0.4914
## F-statistic: 12.11 on 2 and 21 DF,  p-value: 0.0003176
```